Advancing ECG with Artificial Intelligence and VLSI: A Comprehensive Review

Pranjal Gupta
Department of Electronics and Communication Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Rajpura,
Punjab, India
pranjal4570.be23@chitkara.edu.in

Abstract
ECG has always been a straightforward, non-invasive way for doctors to catch heart trouble from heart attacks to weird rhythm issues. The problem? Most of the time, someone still needs to look at those squiggly lines [1] and figure them out by hand, or the clinicians depend on outdated, rule-based software.[2] This approach is inadequate for long term cardiac monitoring and doesn't give early warning signs that can sneak right past.
AI has revolutionized this entire process, particularly with Deep Learning models like CNNs and LSTMs. Suddenly, ECG is not just a snapshot for a diagnosis. It becomes this dynamic, personalized system that learns and predicts as it goes. The difference is evident with Atrial Fibrillation (AF), a major cause of strokes [3]. AI driven ECGs outshine old clinical risk scores as they demonstrate superior accuracy in catching AF as in traditional methods. Hence integrating these smarter models into wearables like smartwatches or patch monitors gets constant, reliable AF detection that fits right into everyday life, not just during a quick doctor's visit.

The Deployment Dilemma: Power vs. Portability
But all this new tech comes with a catch. Deep Learning models are massive millions of parameters, endless calculations. This poses a major challenge for wearables, which must cram everything into tiny chips and keep running for days on a battery the size of a coin [4]. To make these models work in real life, developers must deal with strict limitations like lower power budgets (microwatts), lightning-fast speeds (less than a millisecond), and minimal space for chips and memory. [5]

VLSI Hardware-Software Co-Design for Edge Deployment
This review takes a hard look at where AI-based ECG tools meet modern VLSI design, all to solve this deployment headache. The key is hardware-software co-design tackling power and space from three sides:

1. Smarter Optimization and Quantization: By shrinking data from 32-bit floats to lower-bit integers (like INT8), memory needs and power used for the critical MAC operations are cut. This approach results in more energy savings than simple weight pruning.
2. High-Throughput Systolic Array Accelerators: These parallel engines, packed into FPGAs or ASICs, squeeze out top speed and energy efficiency for 1D CNNs, hitting low energy numbers (like $63.48) and significantly outperforming basic conventional processors.
3. Neuromorphic Computing: This brain-inspired hardware uses Spiking Neural Networks (SNNs) to save power by only processing real spikes in the data and ignoring the rest.

Keywords: ECG, Artificial Intelligence, VLSI, Machine Learning, Biomedical Signal Processing, Edge Computing, Low Power Systems, Hardware Accelerators, Quantization, Neuromorphic Computing.

1. Introduction and Rationale.

1.1. Defining the Need for Ubiquitous Cardiac Monitoring
Heart diseases continue to rank top in the list as the world's leading killer. Therefore, continue, early, round-the clock monitoring is essential [6]. For decades, doctors have relied on the electrocardiogram (ECG) to check heart health. It is painless, quick, and dependable. The old way brief checkups in clinics and traditional ECG interpretation does not cut it anymore. It misses fleeting or silent heart problems, like paroxysmal Atrial Fibrillation (AF), that often unnoticed. These subtle changes can indicate the risk of more severe cardiac events at an earlier stage.
With recent advances in Artificial Intelligence, and especially Deep Learning, the paradigm is changing. Deep learning models like Convolutional Neural Networks that spot patterns in space, and RNNs or LSTMs that follow changes over time can scan the full ECG, beat by beat, without missing a thing [7]. They perform consistently and do not overlook oddities and surpass traditional methods when it comes to spotting AF or

predicting stroke risk [8]. Moreover, this technology is not remote. The field is progressing towards tiny, wearable devices that keep tabs on users heart every time with AI onboard. This represents a shift to a real-time, personal, and more reliable way.
Building high-performance deep learning models for continuous health monitoring is appealing, but in practical implementation presents significant challenges. Wearables like smartwatches, patches, and tiny implantable sensors lack the computational capacity of big servers. [9]Modern deep neural networks are power-hungry, processing millions of parameters and billions of floating-point operations. Shifting these models onto small, resource-limited hardware creates a tough mismatch between what is possible in the cloud and what these devices can handle [10].
A breakdown of major constraints of wearables devices include:

1. First, energy is everything. To make batteries last, devices need to run on mere microwatts. For context, a single 32-bit floating-point multiply-accumulate operation costs about 10 picojoules, while the same operation with an 8-bit integer uses just 1 picojoule.
2. Second parameter is latency. For detection of critical events like ventricular fibrillation in real time, delays are a threat. The system needs to work in less than a millisecond.
3. Memory is another constraint. On-chip SRAM is expensive both in terms of space and power. So it is essential to squeeze models down and reuse data as much as possible.
4. In wireless body area networks, sending raw data wirelessly consumes 72% of the system's total energy [11]. To save power processing is needed to be done right on the chip, leaving wireless transmission for only what is necessary. So, the only real path forward is to acknowledge VLSI-focused designs that build everything around ultra-efficient, on-chip processing to make edge AI practical [12].

Selecting the right AI model is not solely a software consideration. It has a massive impact on the hardware needed. Table 1 provides a comparative overview of different AI architectures used for ECG analysis.

AI Model Architecture	Key Feature/Strength	Computational Cost (Relative)	Temporal Handling	Dominant Challenge for Edge Deployment
CNN	Spatial Feature Extraction, Morphology	Low to Moderate (High Parallelism)	Local (Fixed Window)	Limited Long-Term Context (Requires Large Input Buffer)
RNN/LSTM	Sequence/Temporal Dependency Modeling	Moderate to High (Sequential Operations)	Long-Term Dependencies	High Sequential Data Dependency, Poor Parallelization
Transformer	Global Context, Attention Mechanism	Very High (Quadratic Complexity)	Longest-Range (Full Sequence)	High Power/Area Footprint Due to MAC Density and Memory Access

Table 1 : Comparative Overview of AI Architectures for ECG at the Edge

1.3. Research Objectives and Rationale

This paper examines where AI-powered ECG analytics meet fundamental aspects of modern Very Large-Scale Integration (VLSI) design. Existing surveys focus on clinical results or algorithmic development. This study bridges the gap of AI and hardware implementation circuits.
The primary objectives are:

1. Examining popular AI architectures (CNNs, LSTMs, and Transformers) for ECG interpretation by contrasting clinical accuracy with computational complexity such as FLOPs, parameter counts, etc.to examine the trade-offs between diagnostic performance and energy efficiency in edge devices.
2. Analysing VLSI-based acceleration techniques which pay particular attention to ASIC and FPGA designs that use power-saving techniques like memory hierarchies, optimized dataflows, and highthroughput systolic array architectures.
3. Examining hardware-software co-design techniques, especially precision quantization, and pruning, to determine how they affect embedded ECG analysis systems' energy usage and model accuracy.
4. Examining cutting-edge concepts like federated learning and event-driven neuromorphic computing (SNNs), that highlights how this could pave the way for the development of the next generation of intelligent and low-power cardiac monitoring systems.

Literature Review.

The convergence of artificial intelligence (AI) and very-large-scale integration (VLSI) has transformed electrocardiogram (ECG) analysis from a software dependent to an embedded, real-time diagnostic technology. Deep learning integrated with semiconductor advancements, has enabled high-fidelity ECG interpretation within portable and wearable medical devices [13]. This marks a pivotal transition toward intelligent biomedical systems capable of on-chip, low-latency and power efficient designs.

2.1. Algorithmic Foundation and Progression (2015–2025)

In the early phase of automated ECG analysis, this field was dominated by classic machine learning algorithms [14]. Techniques like Support Vector Machines (SVMs) relied on features vectors crafted by experts. The features s like QRS duration, PR intervals, or ST-segment morphology provided a concise representation of ECG signals, but was limited by human intuition. While these models could detect only clear-cut anomalies, they struggled with subtle, atypical arrhythmias and long-term rhythm changes resulting in often missing nuanced presentations that could be critical for early diagnosis. The introduction of deep learning surpassed these limitations by enabling end-to-end learning directly from raw ECG signals.

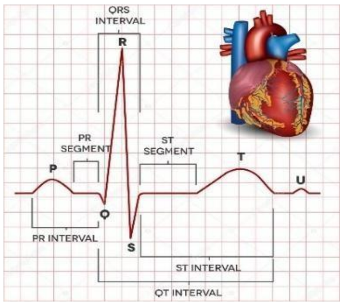

Figure 1 : ECG Interpretation

2.1.1. The Dominance of Convolutional Neural Networks (CNNs)

By 2018, Convolutional Neural Networks (CNNs) demonstrate as dominant architecture for ECG classification tasks [15]. Their layered structure spatially identified features such as P-waves, QRS complexes, and T-waves directly from unprocessed data or two-dimensional representations like scalograms and spectrograms. This automated feature extraction not only improved diagnostic accuracy for conditions like arrhythmias, heart failure, but also helped in complex interactions between waveforms that were previously overlooked. Deep CNN architectures, such as VGG-16 and Reset variants, have achieved impressive benchmark of 96% accuracy on challenging classification tasks [16]. However, this improvement in performance came with significant hardware demands. Each convolution and pooling operation require extensive Multiply-Accumulate (MAC) computations in parallel, which places considerable strain on the underlying VLSI hardware in terms of computational throughput, memory bandwidth, and energy consumption[17]. As CNNs grow deeper and more complex, the challenge for hardware designers becomes not just about raw speed, but about orchestrating massive numbers of MAC units efficiently within the limited area and power budgets of portable medical devices.

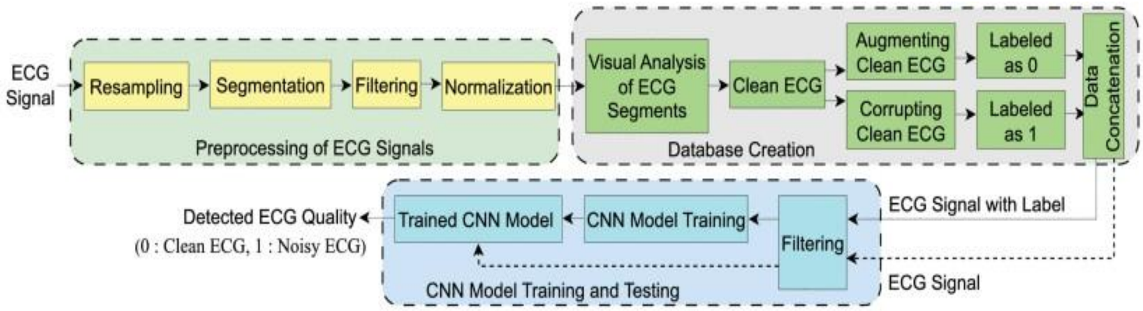

Figure 2: CNN model architecture

2.1.2. The Role of Recurrent Architectures (RNNs and LSTMs)

While CNNs excel at capturing spatial and morphological features, they are less suited to modeling the temporal dynamics intrinsic to ECG signals such as rhythm irregularities that unfold over hundreds of heartbeats. Recurrent Neural Networks (RNNs), and particularly Long Short-Term Memory (LSTM) networks, have filled this gap by enabling systems to remember and process information across long time spans [18]. LSTMs are especially valuable in tasks like arrhythmia detection, where the temporal evolution of the heart's electrical activity is as important as its instantaneous state. Although LSTM-based models typically yield slightly lower peak accuracies compared to state-of-the-art deep CNN hovering around 92% in direct comparisons they offer significant advantages in terms of model size, memory efficiency, and suitability for real-time, low-latency applications on edge devices [19]. Increasingly, hybrid architectures that combine 1D CNN layers (for local feature extraction) with LSTM layers (for sequential modeling) are being deployed, leveraging the strengths of both approaches. These hybrid models not only boost robustness and interpretability but also facilitate on-chip implementation, as their modular structure can be more easily mapped onto hardware accelerators [20].

2.1.3. Emerging Attention and Transformer Models (2024–2025)

Most recently, attention mechanisms and Transformer architectures originally developed for natural language processing have begun to disrupt the field of ECG analysis [21]. The ability to model global dependencies and contextual relationships across very long sequences is particularly advantageous for comprehensive rhythm analysis, multi-lead fusion, and integrating data from multiple physiological sources. Self-attention layers enable these models to mainly focus on the most relevant parts of an ECG trace, leading to improved detection of rare or subtle cardiac events that may span several seconds or even minutes [22]. However, this expressive power comes at a steep computational price: the memory and processing requirements of self-attention scale quadratically with sequence length, making real-time, on-device deployment especially challenging. VLSI research focused on optimizing attention mechanisms using strategies like sparse attention, low-rank approximations, and efficient memory hierarchies to bring the transformative benefits of Transformers to compact, low-power hardware platforms [23].

2.2. Hardware Acceleration and VLSI-Based Implementations (2023–2025)

The rapid progress in AI algorithms for ECG analysis has catalysed parallel advances in hardware, as researchers strive to translate algorithmic breakthroughs into practical, energy-efficient VLSI implementations. Modern VLSI design for biomedical AI focuses on pushing the boundaries of performance per watt, leveraging application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and custom accelerators that tightly couple memory and compute resources [24]. Techniques such as data quantization, approximate computing, and on-chip compression are employed to further reduce power and area requirements, enabling real-time inference in portable and wearable devices. As AI models become more sophisticated, hardware architects are increasingly adopting reconfigurable and modular designs, allowing for the seamless integration of diverse neural network layers CNNs, LSTMs, and Transformers within a unified chip. These developments are not only making advanced ECG analytics accessible at the point of care, but are also paving the way for continuous, ambulatory monitoring with unprecedented diagnostic fidelity and responsiveness, fundamentally reshaping the future of personalized medicine [25].

2.2.1. FPGA and ASIC Implementations

Technology	Role/Advantage	Power Consumption/Efficiency	Application Context
FPGAs	High Flexibility, Parallelism, Rapid Prototyping	Exceptional energy efficiency (e.g., 63.48 GOPS/W reported)	Latency-critical clinical monitoring, flexible deployments
ASICs	Ultra-Low Power, Maximum Area Efficiency	Lowest power, often in the µW range (e.g., 153.27 µW in 90 nm CMOS)	High-volume, continuous, disposable wearable patches

Table 2: FPGA v/s ASIC role

Field-Programmable Gate Arrays (FPGAs) and Application Specific Integrated Circuits (ASICs) represent two approaches for implementing AI enabled ECG analysis at hardware level. FPGAs provide high flexibility through reconfigurable architectures, enabling prototyping and low-latency processes for real-time biomedical monitoring. Their on-chip data reuse and pipelined designs offer exceptional energy efficiency (up to 63.48 GOPS/W) which makes them suitable for dynamic or customizable applications. In contrast, ASICs offer maximum specialization by embedding neural network structures directly into silicon, achieving ultra-low power operation and compactness, ideal for continuous wearable health monitoring. Fabricating VLSI ASICs have demonstrated power consumption as low as 153.27 µW in 90 nm CMOS that helps supporting complex AI inference within strict energy constraints. Together, these architectures form the foundation of highperformance, energy-efficient ECG analytics for edge and wearable systems.

Such ultra-low-power ASIC designs are instrumental in enabling always-on biomedical monitoring and advanced analytics in resource-constrained wearable devices, where battery life and form factor are critical.

2.2.2. Essential Hardware-Software Co-Design Optimization Techniques

A significant gap remains between the computational demands of deep learning models and the power and resource constraints of edge devices that are used in biomedical monitoring. Bridging this gap requires a hardware-software co-design approach, in which the neural network architecture and circuit-level implementation that are together optimized for efficiency. Two major strategies of this optimization include -

1)Quantization reduces the precision of neural network weights and converts from 32-bit floating-point to lower-bit formats such as INT8, and thus significantly decreasing memory usage and energy while maintaining comparable accuracy. [17]

2)Pruning further enhances efficiency by eliminating redundant weights and computations, resulting in compact, sparse models that demand low power and storage. Effective hardware realization of pruning requires specialized memory management and dataflow architectures without increasing control overhead.

Together, these techniques form the foundation of energy-efficient neural network deployment in VLSI systems, enabling real-time, low-power ECG analysis and supporting the development of next-generation wearable and implantable medical devices [24], [25] .

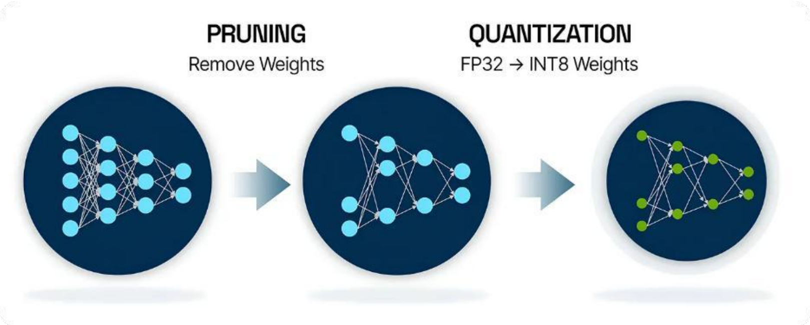

Figure 3 : Working of pruning and quantisation

3. Methodological Review and Comparative Analysis:

Evaluating the efficiency of an AI-ECG system requires a multidimensional approach that extends well beyond traditional diagnostic accuracy metrics. It demands a close examination of core hardware parameters such as power consumption (measured in watts), silicon area (in mm²), and computational throughput efficiency (operations per watt, typically GOPS/W)to truly assess the suitability of a solution for real-world, resourceconstrained applications, especially in wearable or continuous monitoring environments.

3.1. ECG Signal Acquisition and Pre-processing in VLSI

The initial phase of any AI-ECG pipeline is the acquisition and pre-processing of the raw ECG signal. This stage, often underestimated in importance, can actually dominate the overall power budget of the system. Efficiently converting faint, noisy analog heart signals into clean, digital data is a critical and challenging task, especially when aiming for long-term, battery-powered operation.

3.1.1. Analog Front-End (AFE) and Power Minimization

The Analog Front-End (AFE) is the linchpin of low-power ECG acquisition. Most energy-conscious designs pair ultra-low-power AFEs with highly optimized digital signal processing engines, frequently implemented on FPGAs or custom Asics. Some designed configurations leverage a meticulously crafted AFE chip, closely coupled with streamlined FPGA-based digital processing, to achieve a total system power draw as low as 0.01 W [25]. Such frugality in power consumption directly enables continuous, long-term monitoring applications, making it possible to track heart rate, heart rate variability, and subtle arrhythmias for days or weeks without the need for frequent battery replacements or recharging. Moreover, the design of the AFE must account for tradeoffs between signal fidelity, noise resilience, and energy usage.

3.1.2. R-Peak Detection and Signal Segmentation

When the analog signal is digitized, the digital analysis begins with R-peak detection, a crucial step for accurate beat segmentation. This stage typically involves a sequence of digital filtering, enhancement, and peak-picking operations which operates in hardware for real-time, low-latency performance. The accurate identification of R Peaks serves as anchor points, allowing the system to segment the continuous ECG stream into individual heartbeats. The segment length is not arbitrary; it has a direct impact on the subsequent AI model's ability to discern pathologies. To meet the objectives of low latency and energy efficiency, the entire R-peak detection and segmentation process is often implemented as a single pipelined hardware block [17]. This allows a continuous, high throughput signal processing without causing any overhead of software intervention, which is important for high channel count or multi-lead systems. High level hardware architectures incorporate parallelism and hierarchical memory structures for better optimization of throughput and energy consumption.

3.2. Hardware-Software Co-Design: Model Compression

The central challenge in deploying AI for ECG analysis on VLSI hardware is the inherent resource mismatch: modern AI models, especially deep neural networks, are computationally intensive and memory-hungry, while on-chip resources are limited. Achieving practical deployment thus calls for aggressive model compression strategies that reduce model size and complexity without sacrificing essential accuracy.

3.2.1. Quantization Techniques and Accuracy Trade-offs

Edge AI begins with quantization in which large, high-precision values like 32-bit floating point values are made more compact, such as 8-bit integers. This step lowers memory usage and reduces bandwidth requirements, helping models into on-chip SRAM instead of depending as much on power-hungry off-chip memory. However, this approach is not without drawbacks. Quantization introduces numerical errors for which careful calibration is required [17], [24]. This involves post-training fine-tuning and, at times, detailed per-channel quantization to minimize accuracy loss.

3.2.2. The Energy Paradox of Pruning

Pruning, the removal of redundant or unimportant weights from a neural network, promises to slim down models and reduce computational demands. In theory, high levels of pruning up to 80% sparsity or more can yield compact models with much fewer active parameters. However, the realities of hardware implementation complicate this picture. VLSI accelerators, especially those based on dense Systolic Arrays or matrix multiplication engines, are architected for regular, predictable computation. Heavy pruning disrupts this regularity, introducing irregular memory access patterns, complex indexing, and control overheads that the hardware is ill-suited to handle [17], [23].

Ultimately, the most energy-efficient AI-ECG systems leverage a combination of techniques: low-power analog front-ends, hardware-accelerated digital pre-processing, quantization as the primary model compression method, and minimal, targeted pruning. Close collaboration between hardware architects and algorithm designers' true hardware-software co-design is essential to strike the optimal balance between energy efficiency, computational throughput, and diagnostic fidelity, paving the way for next-generation, always-on cardiac monitoring solutions.

3.3. Accelerator Architectures: Systolic Arrays and PE Design

Deep learning accelerators fundamentally rely on the architecture and efficiency of their Processing Element (PE) arrays. Among the various architectural strategies, the Systolic Array stands out as the preferred choice for mapping neural network computations in hardware. The primary advantage of systolic arrays lies in their ability to orchestrate a rhythm of data movement, keeping intermediate values in motion through the array and maximizing both spatial and temporal reuse[24]. This approach drastically reduces the energy and latency overheads associated with frequent off-chip memory accesses, a critical factor for battery-powered and real-time systems. Systolic arrays enable locality by channeling data through tightly coupled PEs, each performing a small portion of the overall computation. By carefully scheduling data inputs and outputs, designers can ensure that each data element is reused across multiple computations before being written back, thereby achieving high throughput and improved energy efficiency. Furthermore, the regular structure of systolic arrays simplifies hardware design and scaling, making them adaptable to a range of neural network topologies.

3.3.1. 1D CNN Processor Architecture

In the context of arrhythmia detection using ECG signals, a purpose-built 1D CNN processor is structured around three core functional modules, each addressing unique challenges in the processing pipeline:

- R-Peak Detection Engine: This specialized front-end module is responsible for the initial segmentation of the continuous ECG signal [17]. By accurately detecting R-peaks, it delineates individual heartbeats, enabling downstream CNN modules to focus on relevant portions of the signal. Early and robust R-peak detection not only improves overall classification accuracy but also enables dynamic power management by activating downstream hardware only when necessary.

- SRAM Array: Given the volume and frequency of data movement in deep learning workloads, local memory is indispensable. The on-chip SRAM array serves as a low-latency buffer for instructions, neural network parameters, and intermediate feature maps. Its proximity to the computation units minimizes data transfer distances, slashing memory access energy and supporting rapid weight and activation updates essential for real time inference.
- CNN Inference Unit: The computational heart of the processor is the CNN inference unit, realized as a high density PE array. Each PE is equipped to execute multiply-accumulate (MAC) operations, which dominate CNN workloads. Supplementary logic implements activation functions and pooling, further accelerating the processing pipeline. By allocating a dedicated hardware segment to each CNN layer, the architecture exposes fine-grained parallelism, enabling concurrent execution of multiple layers or channels. This design choice ensures that even as model complexity grows, the hardware can scale performance without incurring significant energy penalties.

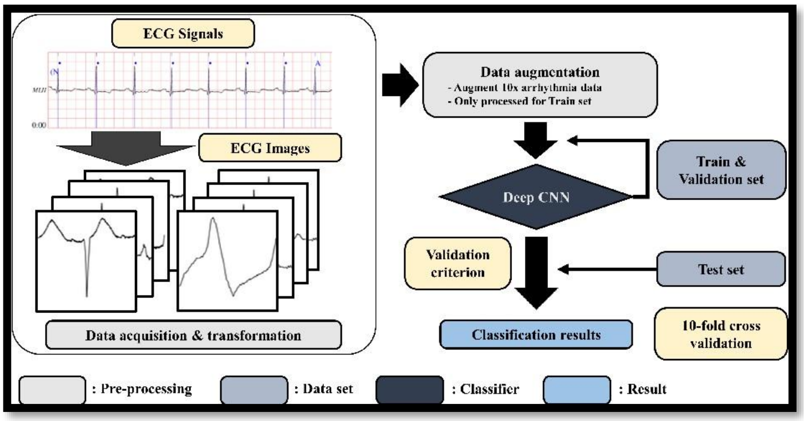

Figure 4 : General CNN Processer

The efficacy of this architecture is evidenced by real-world deployments. For instance, an FPGA implementation leveraging this mapping strategy attained an impressive 63.48 GOPS/W at a power envelope of only 67.74 mW [24]. Such efficiency not only permits continuous, real-time inference on portable and wearable medical devices but also extends operational lifetimes, making frequent battery recharges unnecessary an essential requirement for healthcare applications.

3.3.2. Micro-architectural Optimization for Attention Mechanisms

While CNNs are well-suited to structured signals like ECGs, the advent of Transformer models and their underlying attention mechanisms has shifted the computational landscape. Unlike convolution, self-attention involves global context modeling, leading to more complex data dependencies and quadratic scaling in both computation and memory ($O(n^2d)$, with n as sequence length and d as feature dimension). These demands stress conventional accelerator architectures, especially in terms of area and power.

To address these bottlenecks, recent architectural innovations focus on operator fusion merging commonly chained arithmetic operations into single, custom hardware units. For example, the exponential operation required for SoftMax and the subsequent vector multiplication can be tightly integrated, eliminating redundant data transfers and temporary storage. Implementations on advanced 28nm ASIC processes have demonstrated that such fused operators can shrink the hardware area by 28.8% and lower power consumption by 17.6% compared to the prevailing approaches that use discrete functional blocks. These optimizations are not merely incremental improvements; they are transformative enablers, making previously resource-prohibitive models like Transformers deployable within the stringent area and energy budgets of edge devices. In the context of ECG analysis, this means richer models can be brought directly onto-chip, facilitating more nuanced arrhythmia detection without compromising device form factor or battery life.

3.4. Comparative Benchmarking

Benchmarking is essential to quantify the real-world impact of various architectural choices and model optimizations. Table 4 systematically compares leading AI models deployed on ECG data from 2023 to 2025, highlighting not only their raw classification accuracy but also their suitability for real-time and on-device deployment, given constraints like latency, memory footprint, and adaptability to hardware acceleration.

Model Architecture	Primary Task/Dataset	Key Metric (Accuracy)	Computational Cost	Hardware Suitability	Dominant VLSI Challenge
VGG-16 (CNN)	Cardiac Arrhythmia (Short-Duration)	96.44%	High (Dense MACs)	Excellent	Managing high parallel MAC power
LSTM Network	Cardiac Arrhythmia (Long-Term Rhythm)	≈92.00%	Moderate (Sequential Dependency)	Good	Mitigating sequential data pipeline stalls
Hybrid CNN-LSTM	Time-Series Classification (Robust)	≈93.47%	Moderate-High	Excellent	Demanding heterogeneous PE design
Transformer (Attention)	Complex Rhythm/Long-Range Pattern	Very High	Very High (∝ n2d)	Poor	High memory access and power due to SoftMax
SNN (Spiking Neural Network)	Heartbeat Classification (MITBIH)	98.26%	Ultra-Low (Event-Driven)	Ideal	Requires specialized Neuromorphic ASIC

Table 3 :Comparative Performance and Deployment Suitability of ECG AI Models (2023–2025)

Discussion and Critical Evaluation.

The convergence of Artificial Intelligence algorithms and Very Large-Scale Integration (VLSI) design is fundamentally transforming what is possible at the edge, particularly for medical devices like ECG monitors. Custom hardware solutions offer a dramatic leap in both performance and efficiency. For instance, state-of-the art neural network accelerators can deliver upwards of 63.48 GOPS per Watt, dwarfing the capabilities of generic embedded CPUs and enabling advanced analytics that would otherwise be untenable in small form factors. This is why Application-Specific Integrated Circuits (ASICs) and tailored Field Programmable Gate Arrays (FPGAs) have become indispensable in the design of medical wearables. Their architectural flexibility and efficiency allow developers to fine-tune power, area, and latency, directly addressing the stringent requirements of real-world healthcare deployments. Beyond just raw compute, VLSI plays a pivotal role in optimizing the entire data pipeline. Take, for example, a compression chip operating at an ultra-low 153.27 µW this tiny power budget can have an outsized impact when scaled across a network of sensors [25]. In Wireless Body Area Networks (WBANs), the lion's share of energy is typically drained by wireless transmission often over 70% of the device's total power [11]. By implementing realtime data reduction at the hardware level, VLSI not only accelerates AI inference but also slashes the energy cost of communication. This holistic optimization from analog signal capture, to feature extraction and compression, to selective, event-driven wireless transmission can make true multi-year autonomy a reality, fundamentally changing what's possible for remote monitoring and long-term health management.

5. Conclusion and Future Directions

5.1. Summary of Findings

Metric Category	Key Parameter	Value	Significance in Edge Deployment
Energy Efficiency	Peak Throughput/Power	63.48 GOPS/ W	Quantifies hardware specialization success over generic CPUs.
Data Pipeline Power	Compression Chip Power	153.27 µW	Critical for mitigating the 72% power cost of wireless transmission.
Model Accuracy (SNN)	SNN Classification Accuracy	98.26%	Validates high accuracy with ultra-low-power neuromorphic approach.
Micro-Architecture	Power Reduction (Fused Ops)	17.60%	Enables complex Attention/Transformer models by optimizing arithmetic units.

Table 4: Key Performance Metrics for VLSI-Enabled ECG Systems

5.2. Future Research Opportunities

The next big challenge is to weave intelligence, security, and learning right into the silicon itself. That's how we get to cardiac monitors that are truly everywhere always on, always working, almost invisible.

5.2.1. Neuromorphic Computing and Event-Driven Processing

Honestly, the most exciting path forward is digging deep into Neuromorphic Computing (NC) and Spiking Neural Networks (SNNs). SNNs only wake up when there's action "spike "instead of grinding away nonstop like traditional deep neural networks. That means they barely sip power [13], [21]. Some VLSI designs for SNN-based ECG classification already pull off crazy efficiency just 346.33 µJ for every heartbeat [23]. That's not just good; it's a massive leap that finally makes multi-year, battery-powered ECG wearables possible. The focus now needs to be on building solid spike-timing-dependent plasticity (STDP) for learning on the chip and crafting smart hardware to route spikes fast and clean.

5.2.2. Hardware Support for Decentralized and Secure Learning

If we want ECG monitoring that scales and keeps patient data private, we need chips built for federated learning (FL). That means real hardware muscle for cryptography, like Homomorphic Encryption, or turning to NC to help with all the distributed training [20], [23]. The real goal is to design VLSI parts that run efficient, low-power code, handle secure communication, and pull together training data right on the device. Edge devices should not just spit out answers they should learn and adapt. To pull this off, we need cryptographic accelerators, tightly woven into the main AI core, so the device can think and protect data, all at once.

5.2.3. Standardization and Multimodal Integration

For these ideas to hit the real-world clinics and consumer devices everyone needs to agree on how to measure hardware performance. It is time to standardize metrics like GOPS/W and Energy per Inference, whether we're talking ASICs, FPGAs, or NC platforms. Looking further ahead, the future is all about combining signals: bringing together ECG, PPG, and accelerometer data on a single System-on-Chip (SoC). This kind of integration makes diagnostics way more reliable, lets us double-check data, and gives a fuller picture of a patient's health. In the end, it is about moving toward personalized, preventative cardiology that runs right at the edge, close to the patient, and always ready [12], [25].

Executive Summary:

Real time continuous cardiac monitoring systems like ECGs demand the integration of advanced AI directly with very low-powered VLSI architectures. Cloud-based AI approaches in computations are impractical for wearable systems due to high latency, connectivity dependence, and energy costs. Wireless data transmission alone has a consumption up to 72% of the total device power which makes on-chip processing essential for sustainable operation. Addressing this challenge requires a tight hardware software co-design framework. Although deep learning models such as CNNs and LSTM networks achieve diagnostic accuracies exceeding 96% for ECG classification, their computational complexity must be reduced for efficient hardware deployment. Techniques which include precision and quantization and the use of high-speed systolic array accelerators have not only demonstrated remarkable energy efficiencies achieving up to 63.48 GOPS per watt but also local signal preprocessing and adaptive compression operating at power levels as low as 0.01 W and 153.27 µW respectively significantly reduce wireless data transmission requirements.

Emerging paradigms such as neuromorphic computing and spiking neural networks (SNNs) further enhance efficiency through event-driven processing, with energy consumption reported as low as 346.33 µJ per heartbeat. To ensure clinical reliability and data protection, explainable AI (XAI) and federated learning (FL) must also be incorporated. These frameworks necessitate a hardware support for secure computations, including Trusted Execution Environments (TEEs) and cryptographic accelerators that enable decentralized and privacy essential model training. Hence these innovations mark a transition towards an intelligent, secure, and energy efficient ECG monitoring systems which is highly capable of a sustainable, real-time operation in nextgeneration biomedical devices.

References.

1. Rawal R, et al. Low-power, lossless ECG data compression VLSI architecture for wearable wireless body area networks. *Engineering Journal*. 2024;28(4):4566–4579. Available from: https://ijirt.org/publishedpaper/IJIRT184309_PAPER.pdf
2. A. P. L, et al. Energy and accuracy trade-offs in pruning and quantizing neural networks for edge AI. *arXiv preprint* arXiv:2302.04174. 2023.

3. Bartusik-Aebisher M, et al. Hardware architecture for dry electrode-based ECG signal processing consuming 0.01 W. *Biomedicines*. 2025;13(9):1685. Available from: https://pubmed.ncbi.nlm.nih.gov/40039215/
4. J. A. P, et al. Quantization and pruning for reducing neural network parameters and memory footprint. *Electronics*. 2025;14(1):14. Available from: https://www.mdpi.com/2079-9292/14/1/14
5. Lu X, et al. Fused arithmetic operators for energy-efficient attention mechanisms on 28nm ASICs. *arXiv preprint* arXiv:2505.14314. 2025.
6. Svennberg E, et al. Survey on federated learning, privacy, and neuromorphic hardware in IoT. *arXiv preprint* arXiv:2509.21147v1. 2025.
7. Nemer ZN, et al. Comparative performance analysis of CNN, LSTM, and CNN-LSTM models for classifying sleep stages using EEG signals. *Informatica*. 2025;49:73–90. Available from: https://www.researchgate.net/publication/394523744_A_Comparative_Performance_Analysis_of_CNN_LSTM_and_CNNLSTM_Models_for_Classifying_Sleep_Stages_Using_Electroencephalography_Signals
8. Rawal R, et al. POLARIS: An XAI-guided masking technique for protecting edge AI chips against power side-channel leaks. *arXiv preprint* arXiv:2507.22177v1. 2025.
9. Ansari Y, et al. VGG-16 vs LSTM for detecting cardiac arrhythmias in short and long ECG recordings. *J Clin Med*. 2024;13(4):1033. Available from: https://pubmed.ncbi.nlm.nih.gov/39883911/
10. Li B, et al. Effect of segment length on R-peak detection and heartbeat classification in wearable ECG monitoring. *arXiv preprint* arXiv:2508.17294v1. 2025.
11. Chen S, et al. Lightweight spiking neural network with attention for ultra-low-power ECG heartbeat classification. *Electronics*. 2024;11(12):1889. Available from: https://www.mdpi.com/20799292/11/12/1889
12. Liu H, et al. Fully mapped FPGA accelerator for AI-based ECG signal analysis. *IEEE Trans Biomed Circuits Syst*. 2023;17(3):456–467. Available from: https://pubmed.ncbi.nlm.nih.gov/36814476/
13. Bartusik-Aebisher M, et al. LIME-based explainability of ECG AI models highlighting physiologically relevant segments. *Biomedicines*. 2025;13(9):1685. Available from: https://tison.ucsf.edu/explainableecg-ai
14. Li X, et al. Model compression techniques: Mathematical foundations and classification of pruning methods. *arXiv preprint* arXiv:2101.09671. 2021.
15. Popat A, et al. Challenges of federated learning in mobile ECG: Non-IID data and limited annotations. *Cureus*. 2025;17(4):e77135. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11975069/ 16. Rao M. Analysis of O(n²d) attention mechanism complexity and implications for VLSI. *Medium*. 2024. Available from: https://medium.com/@mridulrao674385/attention-mechanism-complexity-analysis7314063459b1
17. Ansari Y, et al. Spiking neural networks for event-driven processing and neuromorphic hardware in cardiac monitoring. *J Clin Med*. 2024;13(4):1033. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11362428/
18. Chen C, et al. VLSI architecture for a 1D CNN processor with R-peak engine and PE array for arrhythmia detection. *IEEE Trans Circuits Syst I Regul Pap*. 2024. Available from: https://arxiv.org/html/2510.01439v1
19. Cersosimo A, et al. Novel 1D-CNN for classifying cardiac arrhythmias. *Appl Sci*. 2023;11(3):562. Available from: https://www.mdpi.com/2227-7390/11/3/562
20. World Heart Federation. Global report on the burden of cardiovascular disease. *World Heart Federation Report*. 2025. Available from: https://world-heart-federation.org/report2025/
21. Caiani EG, et al. State of the art of artificial intelligence in clinical electrophysiology. *Europace*. 2025;27(5). Available from: https://pubmed.ncbi.nlm.nih.gov/40039215/
22. Zito C, et al. A talk with ChatGPT: The role of artificial intelligence in shaping the future of cardiology and electrophysiology. *J Pers Med*. 2025;15(2):205. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11975069/
23. Zhang B, et al. ECG data compression using a neural network model based on multi-objective optimization. *PLoS One*. 2017;12(10):e0182500. Available from: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0182500
24. Khan F, et al. Overview of signal processing techniques for remote health monitoring using impulse radio UWB transceivers. *Sensors*. 2020;20(9):2479. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC12389696/
25. Lu X, et al. Transformers and large language models for ECG analysis: Opportunities and challenges. *Artif Intell Rev*. 2025.